

DP Computer Science: A 1.1.7 Secondary Memory

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

1. Describe what separates **internal** from **external** secondary storage—that the distinction is **location and connection**, not what the device stores
2. Describe the three internal types (**HDD, SSD, eMMC**), giving for each the technology it uses, one or two defining characteristics, and a typical use
3. Describe the external types (**external HDD and SSD, optical drive, flash drive, memory card, NAS**) in the same terms: technology, characteristic, use

Command Term Note: Each intention lands at **Describe**—a detailed account. Naming a device is not describing it, and the lesson does not ask students to evaluate or compare beyond the internal and external split.

Key Vocabulary (Tier 3)

Term	Definition
Secondary storage	Non-volatile storage that holds data whether the machine is on or off
Internal storage	Fitted inside the computer case, connected directly to the motherboard
External storage	Connected from outside the case through an interface such as USB or a network
Interface	The connection through which an external device links to the computer, e.g., USB
Flash memory	Storage with no moving parts, used by SSDs, eMMC, flash drives, and memory cards
NAS (Network Attached Storage)	A storage device connected to a network so many machines can reach it

Pedagogical Focus

- **Thinking Skills (ATL):** The skill students consciously practise is **transfer**—taking the organising principle built from the taught types and applying it to a type they have never met. Name this at the final task: *"You are about to be handed a device we have not studied; you can still place it, because the classification tells you what to look for—location and connection."*
- **Conceptual Understanding (ATT):** Teach this explicitly and in order, building one organising idea rather than eight separate facts. The test of whether the concept has formed is not whether students can recite the eight types, but whether they can place a ninth they were never taught.

Secondary ATLs present: Communication, through the written describe answers using the sentence frame.

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three short recall items from A1.1.4 (Primary Memory):

Question	Source
1. Which type of primary memory is volatile and loses its contents when the power is switched off?	[A1.1.4]
2. A1.1.4 said the permanent versions of what is in RAM "live in storage." Is that storage volatile or non-volatile?	[A1.1.4]
3. State one reason ROM rather than RAM holds a computer's startup instructions	[A1.1.4]

Note: A1.1.4 is the immediately prior lesson, and it ended on the idea that RAM's permanent copies live in storage. That storage is today's topic, so it is the retrieval bridge.

Part 2: Main Sequence — ~60%

Phase A: Concrete Example and the Organising Principle (~6 mins)

Start from One Clear Pair:

Device	Classification	Technology
SSD soldered inside a laptop	Internal	Flash memory
Same SSD in an enclosure on a USB cable	External	Flash memory

Draw Out the Principle:

What makes storage internal or external is **where it sits** and **how it connects**, not what it stores or how permanent it is.

Two Questions Students Will Now Test Against Every Type:

1. Is it inside the case with a direct connection to the motherboard?
2. Does it therefore trade portability for speed?

Correct the Central Misconception Here, Before Any List:

- Internal/External is **not** Primary/Secondary
- All secondary storage is **non-volatile**

Phase B: Progressive Abstraction Across the Taught Types (~8 mins)

Task: Students draw a four-column table and place each type themselves by applying the principle from Phase A—not copying a completed table.

Type	Internal or External?	Technology	Typical Use
HDD	Internal	Magnetic, moving parts	Cheap per gigabyte; capacity over speed
SSD	Internal	Flash, no moving parts	Fast; holds the OS
eMMC	Internal	Flash soldered to the board	Compact and cheap; built-in storage of phones and tablets
External HDD	External	Magnetic, moving parts, in enclosure	Portability; backup
External SSD	External	Flash, in enclosure	Portability with speed
Optical Drive	External	Optical (laser)	Reading/writing CDs, DVDs, Blu-rays
Flash Drive	External	Flash	Portable file transfer

Type	Internal or External?	Technology	Typical Use
Memory Card	External	Flash	Portable storage for cameras, phones
NAS	External (network)	HDD or SSD in network enclosure	Shared storage across multiple devices

The table students build is the abstraction made visible. The eBook is not used here; it is set as reading around the lesson.

Phase C: Transfer Task — The Test of the Principle (~4 mins)

Task: Hand students one device they were never taught.

 **Explicit Flag:** *"This is not a storage type you need to know or will be examined on. It is here only to test whether the classification works on something new."*

Workable Vehicle: LTO Tape Cartridge

Description	Details
Technology	Magnetic tape in a cartridge
Connection	Read by a separate tape drive
Characteristics	Very high capacity, very slow
Use	Archive backup

Student Task: Place it—internal or external—and justify from the principle.

Expected Reasoning:

It connects to the computer through a separate drive and interface rather than sitting on the motherboard, so it is **external**. Its slow, high-capacity, archive profile fits external backup.

The point is not the tape; it is that the principle placed it.

Part 3: Deliberate Practice — ~20%

Activity: Individual, at the command term, on the taught types only.

Question 1:

"Describe two differences between internal and external secondary storage." [2 marks]

Question 2:

"Describe the characteristics of an SSD and state one reason it is preferred to an HDD for storing an operating system." [3 marks]

Sentence Frame:

"[Device name] is [internal/external] storage because it [location/connection]. It uses [technology] technology, which means it [characteristic]. This makes it suitable for [typical use]."

Protect this phase if time runs short; it is where each student proves the Describe form for themselves.

Part 4: Exit Check — ~10%

One Describe Question (3 marks):

"A smartphone has no drive bay and nothing plugged into it, yet it keeps thousands of photos through every restart. Name the internal storage type it most likely uses, and describe it: the technology, one characteristic, and why it suits a phone."

Mark Scheme:

Marks	Criteria
[1]	Names eMMC
[1]	Describes technology: flash memory soldered directly to the board
[1]	Gives characteristic and reason: compact and cheap/low power suits a phone because there is no room for a separate drive and it keeps cost down

Teacher Note: This checks both the Describe form and, through "internal," that the classification held. A student who names eMMC but gives no reason has stopped short of Describe and needs the "and why" prompt next lesson.

3. Summary: Internal vs. External Storage

Internal Storage

Type	Technology	Characteristics	Typical Use
HDD	Magnetic, moving parts	Cheap per gigabyte; high capacity; slower	Mass storage, archives
SSD	Flash, no moving parts	Fast; reliable; more expensive per GB	Operating system, frequently used programs
eMMC	Flash soldered to board	Compact; cheap; lower performance than SSD	Smartphones, tablets, budget devices

External Storage

Type	Technology	Characteristics	Typical Use
External HDD	Magnetic, moving parts	Portable; high capacity; interface-limited	Backup, file transfer
External SSD	Flash	Portable; fast; more expensive	High-speed backup, video editing
Optical Drive	Optical (laser)	Read-only or write-once; large capacity per disc	Media distribution, archival
Flash Drive	Flash	Small; portable; inexpensive	File transfer, portable programs
Memory Card	Flash	Compact; removable	Cameras, phones, portable devices
NAS	HDD or SSD in network enclosure	Shared access; centralised storage	Home/office storage, media servers

4. Differentiation & Extension

Support (Scaffolds That Can Be Removed)

Support Strategy	Description
Summary Table (Partial)	Type names and internal/external column already entered; students add only technology and use
Sentence Frame	On the desk for written answers
Transfer Task Prompts	Two-question prompt: "Where does it connect? Inside on the motherboard, or outside through an interface?"

Challenge (Deeper or Wider, at the Command Term)

Challenge Strategy	Description
Second Unfamiliar Device	Give a UFS chip in a modern phone—place as internal, flash soldered to the board, same reasoning as eMMC. Deepens transfer without adding examinable content.

Beyond the Statement (Flag as Such)

Topic	Location in Syllabus
RAID, SAN	Beyond this statement
Tape as a taught type	Not required
Backup strategies	Not required
Cloud storage	A1.1.9
Emerging storage (3D NAND, DNA, quantum)	Beyond this statement

Note: The tape cartridge and UFS chip appear in this lesson only as transfer vehicles for the classification, **not as content students must know**.

5. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

Activity	Description
"Mystery Box"	Students examine various storage devices and speculate about their function and characteristics
"Storage in the News"	Discuss recent technology trends related to storage
"Storage Device Dissection"	Safely dissect an old HDD to show internal components (platters, read/write heads)
"Storage Technology Research"	Groups research a specific technology and present findings
"Storage Spectrum"	Place different technologies on a spectrum from "Fastest/Most Expensive" to "Slowest/Least Expensive"
"Storage Solution Design" Challenge	Design appropriate storage solutions for real-world scenarios (family photos, computer backup, gaming server, media centre)
"Debate: The Future of Storage"	Teams debate continued dominance of local storage vs. rise of cloud storage

6. Learning Objectives (Summary)

Knowledge and Understanding

- Define secondary memory and differentiate between internal and external types

- Identify and describe various types of internal secondary storage (SSD, HDD, eMMC)
- Identify and describe various types of external secondary storage (external SSD/HDD, optical, flash drives, memory cards, NAS)
- Explain typical usage scenarios for each type of storage device

Skills

- Compare and contrast characteristics (speed, capacity, cost, portability) of different storage technologies
- Analyse real-world scenarios and recommend appropriate storage solutions
- Evaluate trade-offs between different storage options

ATL Skills

Skill	Description
Communication	Effectively communicate understanding through discussions and written explanations
Collaboration	Work collaboratively to analyse scenarios and design storage solutions
Critical Thinking	Analyse advantages/disadvantages; evaluate suitability for various applications
Information Literacy	Use online resources to deepen understanding
Reflection	Reflect on own use of storage devices

7. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A 1.1.7 with quiz)
- **Quizlet** (Vocabulary flashcards)
- **Worksheet** (Summary Table)

Author's Reflection

"This is the second of three conceptual-understanding lessons in the block. The organising idea for the lesson is that internal versus external is decided by location and connection, not by what a device stores or how permanent it is—which is exactly the distinction students collapse when they confuse it with primary versus secondary. So this lesson teaches that explicitly.

The lesson ends by handing students a device they were never taught and asking them to place it. A classification you can only apply to the examples you were given is not really a classification, so the transfer step is the real test of whether the concept landed. The device itself is not examinable and I have flagged it as such throughout; it is there only to prove the principle travels."